

processing speed.

In fact, dealing with this type of speed difference usually requires implementing delaying measures, such as inserting a number of wait cycles to keep pace with the flash-memory's slower access speed. Ultimately, this type of 'workaround' solution produces a disappointingly low limit in the microcomputer's overall performance and wastes valuable design potential.

To counter this situation, Renesas has developed a high-speed flash-memory technology that can operate at a maximum speed of 100 MHz. Based on a MONOS (metal-oxide-nitride-oxide-silicon) structure, it has the fastest access speed of any flash memory successfully commercialised to date. The RX600 series of microcomputers adopted the MONOS technology for internal flash. As a result, RX600 microcomputers can perform single-cycle memory access at up to 100 MHz with no wait state.

90nm fabrication process with low power consumption

RX devices use the proven 90nm process for manufacturing economical microcomputers. The primary advantages of this process are higher speeds due to lower capacitive loads (in gates and wiring) and reduced current consumption in normal operation.

Two other chip-design factors also reduce power consumption significantly. The first is a special clock-gating circuit which saves power in normal operation by turning off the clock supply to logic that is not being used, whenever such a shutdown is appropriate.

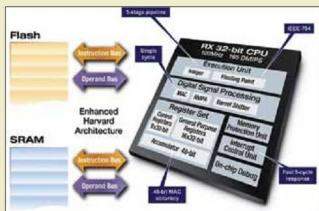
The second is a leakage-current minimisation design technique that uses multiple thresholds. It's generally true that increasing the threshold voltage (V_{th}) of a circuit element reduces leak current and lowers power consumption.

The downside of this approach, though, is that it makes high-speed operation more difficult. For the RX microcomputers, the basic circuit design uses a two-step approach. First, RX uses low-speed cells (high- V_{th} , low-leakage types) to keep leakage current low during normal operation. The high-speed operation is assigned to critical-path cells (a fast, low- V_{th} design). The result of this critical-path analysis and optimisation design approach is measured by the excellent power efficiency of RX microcomputers, which achieve an overall power consumption of 0.5 mA per MHz.

Extreme connectivity

The combination of Ethernet, USB and CAN implemented on a single chip is a key success factor, given today's ubiquitous demand for connectivity. This feature, combined with the powerful RX core of 165 DMIPS performance at 100 MHz, aims to support the growing worldwide demand for networking capabilities. The first chips in this new product line to realise this objective are the microcomputers in the RX62N and RX621 groups.

Both these support the USB 2.0 standard. Those in the RX62N group also incorporate an Ethernet controller, with CAN (controller area network) capability available as an option. By contrast, CAN is a standard feature of the microcomputers in the RX621 group. Some models have dual USB 2.0 channels, in



100MHz, 165DMIPS RX 32-bit CPU structure

which each channel can be set independently to either host or slave mode.

RX platform product line-up

The current RX product line consists of the RX610, RX62N, RX621 and RX62T groups. Of these, the devices in the RX610 group with up to 2MB Flash and 128kB RAM are ideal for applications such as high-end office equipment and HMI terminals that must run large-scale programs rapidly at high performance levels.

The microcomputers in the RX62N and RX621 groups provide enhanced connectivity. The RX62N group is equipped with an Ethernet controller, USB 2.0 Full-Speed (Host, Function, OTG) and CAN (selectable as an option), while the RX621 group is equipped with USB 2.0 Full-Speed (Host, Function, OTG) and CAN. The main features of the RX62N and RX621 groups are compatible as usual for Renesas RX platform.

By contrast, those in the RX62T group offer enhanced motor-control functions. These include a Multifunction Timer Pulse Unit 3 (MTU3), which can drive up to two 3-phase motors, and a 16-bit General PWM Timer (GPT), which can drive a single three-phase motor. As a result, the RX62T products can drive up to three 3-phase motors at the same time with a single chip. Improved analogue functions for system cost reductions and improved ease of use in motor vector control applications have also been implemented in the new RX62T family. It features two 12-bit analogue-to-digital converter (ADC) units with operation amplifier and one 10-bit ADC unit, and can capture analogue input values from up to 20 channels at a minimum conversion time of 1 ms.

Supporting tools

The RX600 series comes with a JTAG debugger interface which enables connection of Renesas E1 or E20 on-chip debugger or similar JTAG third-party systems. Various Renesas starter kits are also available that allow developers to quickly evaluate the potential of RX products to upgrade their existing designs while improving the performance, functionality and cost of their systems.

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